

Open Source as Institutional Infrastructure

Lessons from the UC OSPO Network

Data Science Center, UCLA Library

UCLA OSPO Lead, UC OSPO Network

The invisible layer

Research data workflows run on open source:

- Data collection, cleaning, analysis, visualization
- Reproducibility tooling: containers, workflows, version control
- Repositories and scholarly infrastructure

92% of researchers use software; **67%** say their research would be impossible without it.

The people keeping it running are increasingly called

Research Software Engineers (RSEs), a growing professional community with no standard institutional home.

If every organization had to replace open source with proprietary equivalents, the cost would exceed **\$8.8 trillion**.

Hettrick et al. (SSSI, 2014/2022) ·
Hoffmann, Nagle & Zhou (2024)
doi:10.2139/ssrn.4693148



Jupyter



Python



R



Git



Spark



Pandas



NumPy



Docker



Ray



ggplot2



dplyr



Snakemake



Nextflow

We've been here before

People in this room helped build the open data movement:

- **FAIR** principles: from aspiration to institutional expectation
- Data management plans embedded in grant workflows
- Curation, provenance, and stewardship as professional services
- Open science infrastructure: repositories, PIDs, metadata standards
- **FAIR4RS**: FAIR principles now extended to research software

Software is the next frontier, and UC has been showing the way for 50 years: **BSD**, Unix, Ceph, Apache Spark, **RISC-V**, Ray. The pipeline from university research to global infrastructure is not rare or accidental.

“The research done in academia was the seed. Foundations provided the soil. But the community was the sun.”

Barker et al. (2022) doi:10.1038/s41597-022-01710-x ·
Ruff, N. (2026) “The Role of Foundations in Advancing

Open Collaboration and Innovation,” UC Open Summit.
youtu.be/eBriL3CDNeo

The UC research-to-infrastructure pipeline

UC Berkeley |



Jupyter



Apache Spark



RISC-V



Ray



BSD Unix

UC Santa Cruz |



Ceph

UCSC Genome Browser

UC San Diego |

Cytoscape

EEGLAB



UCSD Pascal

UCLA |



Processing



Named Data Networking

tz database

UC San Francisco |

ChimeraX



OpenMM

UC Davis |



sourmash

ERPLAB

Summer 2000. UC Santa Cruz.

PhD student Jim Kent assembled the human genome on 50 borrowed PCs, icing his wrists to keep coding. He finished **three days** before a corporate competitor's supercomputer.

The genome stayed in the public domain.

Kent, J.W. (2002). BLAT - The BLAST-Like Alignment Tool. *Genome Research* 12, 656–664. · genome.ucsc.edu

What is an academic OSPO?

An Open Source Program Office

supports, governs, and promotes open source within an institution.

- Originated in industry:
~77% of large tech companies have one
- First academic OSPO:
Johns Hopkins, 2019
- **34 academic OSPOs** in CURIOSS worldwide and growing
- Core functions: policy, licensing guidance, best practices, education, community



● UC OSPO Network · ● CURIOSS member

World map showing 34 CURIOSS member institutions. Six U

The UC OSPO Network

A **multi-campus collaboration** treating open source as shared infrastructure.

- 6 campuses: Santa Cruz, Berkeley, Davis, Los Angeles, Santa Barbara, San Diego
 - Launched April 2024, funded by the Alfred P. Sloan Foundation
 - **\$1.85M** in external funding to date
 - Lead: UC Santa Cruz, the first OSPO in a large state university system
 - Serves 280,000+ students and 25,000+ faculty
- Acts as a **neutral convener**: no single campus, company, or grant

cycle can pull the work
away.

Ruff (2026) UC Open
Summit .

youtu.be/eBriL3CDNeo



Map of California showing six UC OSPO Network campuses:

Three thematic areas



Discovery

Mapping who
does what
across the UC
system:
repos,
contributors,
tools, and
practices



Sustainability

Keeping
projects and
communities
healthy
through

licensing,
project health,
and
community
support



Education

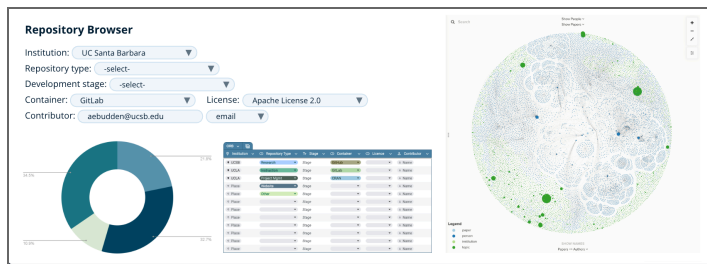
Coordinated
training and
learning
pathways
across
campuses

Discovery: knowing what's there

The 4 Ps framework,
applied at system scale:

- **People:** who contributes to OS software at UC?
- **Product:** what has been built?
- **Practice:** how do faculty and staff engage?
- **Perception:** what are contributors' experiences?

Tools: GitHub pipeline
(200,000+ repos scanned;
~52,000 institutionally
affiliated across 10
campuses), network-wide
survey (294 respondents),
UC Open Repository
Browser (UC ORB)
Gomez et al. (2025)
arxiv:2506.18359 .
Scarlett et al. (2025)
doi:10.31235/osf.io/p8bx6_v1



UC Open Repository Browser showing filter controls, repository type breakdown, data table of UC campus repos, and a network visualization of project relationships

Sustainability: keeping it running

58% of UC open source contributors are also maintainers. Not users: stewards.

What they asked for most: sustainability grants, computing infrastructure, learning communities.

#1 challenge: finding time to write documentation.

Nationally: **60%** of maintainers are unpaid; **43%** report burnout.
Network services:

- Cross-campus licensing working group (UCOP + tech transfer + IT)
- Project health assessment (OSSPREY)
- Containerization support · Community management

Scarlett et al. (2025)

doi:10.31235/osf.io/p8bx6_v1 ·

Tidelift (2024) State of the Open Source Maintainer Report

58%

of UC OSS contributors
are also **maintainers**
not just users, stewards

60%

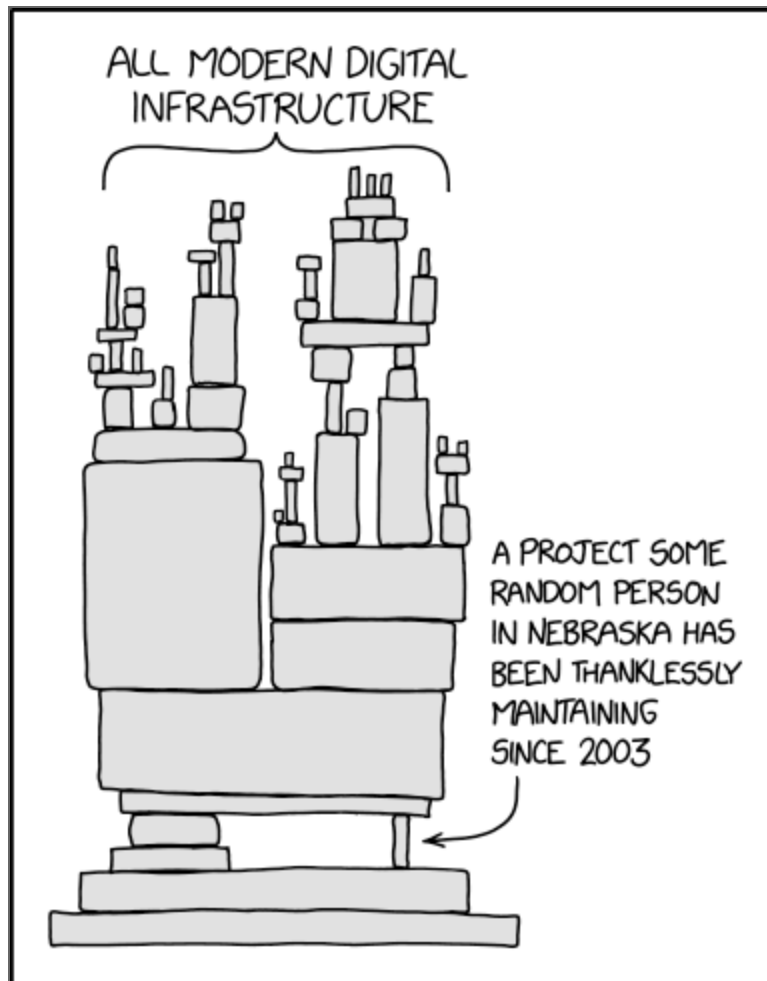
unpaid nationally

43%

report burnout

Tidelift (2024) State of the Open Source
Maintainer Report

The dependency problem



XKCD #2347 'Dependency': a towering stack of blocks labeled 'All modern digital infrastructure' balances on a single tiny block labeled 'A project some random person in Nebraska has been thanklessly maintaining since 2003'.

By Randall Munroe.

xkcd #2347 (CC BY-NC 2.5) · UC survey: 58% of contributors are maintainers · Tidelift (2024): 60% unpaid, 43% burnt out

AI and open source: a new pressure

AI code generation is changing what it means to maintain a project.

- **Slop PRs:** AI-generated contributions that look complete but are off-spec, shallow, or carry subtle bugs
- Review burden grows while maintainer capacity stays flat
- The OSS projects that trained the models are now flooded by their output
- Security risk: AI-generated code in dependencies with undetected vulnerabilities

OSPOs are developing

responses: contribution disclosure policies, AI-aware code review norms, project-level guidance.

Baltes, Cheong & Treude (2026)
arxiv:2603.27249 · GitClear (2024)
“Coding on Copilot”

The structural shift

70% of new AI PhDs go straight to industry
(a decade ago: 50/50)

90% of notable AI models come from a handful of companies

Academia is being sidelined from the systems it helped create.

Ruff (2026) UC

Open Summit ·

youtu.be/eBriL3CDNeo

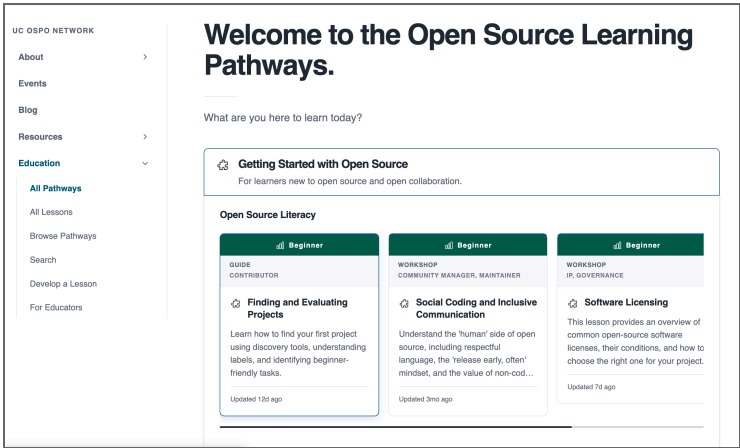
Education: training as infrastructure

Gap analysis →

curriculum inventory →

learning pathways

- Inventoried existing materials: Carpentries, CodeRefinery, The Turing Way, UC-specific content
- Organized by topic (licensing, sustainability, community) and by role (contributor, maintainer, manager)
- Published at **ucspo.net/education**
- Coordinated with data literacy programs and Carpentries infrastructure



UC OSPO Network Education site showing the Open Source Learning Pathways, organized by role, skill level, and pathway, with beginner lesson cards including Software Licensing, Social Coding, and Finding and Evaluating Projects

Distributed expertise, system-wide capacity

Each campus brings what others don't. The network routes it to where it's needed.

Each campus contributes

- Unique domain expertise and staff knowledge
- Local relationships with researchers and faculty
- Pilots that can only run with campus context
- Specialized capacity others can learn from

The network multiplies it

- Expertise flows across all six campuses
- Shared staffing no single campus could fund
- Smaller campuses build capacity they couldn't alone
- A success at one becomes a template for five

Governed by an OSPO Leadership Group and thematic working groups, one per campus, coordinated at system level.

Lessons from two years

What the network model enables that single campuses cannot:

- .. **Shared staffing:** community manager, licensing specialist, technical roles
- 2. **Policy leverage:** the network has standing with UCOP; individual campuses do not
- 3. **Data at scale:** the GitHub and survey analyses require system-wide scope to mean anything
- 4. **Equity:** smaller campuses get services they could not build alone
- 5. **A replicable model:** \$1.85M produced something other state systems can follow

Where this meets your work

For data professionals, the OSPO sits at the intersection of:

- Research software engineering ↔ data management
- Licensing compliance ↔ open data policy
- Reproducibility tooling ↔ FAIR / FAIR4RS
- Training infrastructure ↔ data literacy programs
- RSE community ↔ library research support

For your institution:

- Where does open source software governance currently live?
- What would it take to treat it as infrastructure rather than a side project?

You're already doing this work

If you do technical data services, researchers are already bringing you these questions:

- “What license should I use for this code?”
- “Is this library still maintained? Should I depend on it?”
- “How do I make this reproducible and citable?”
- “My funder wants a software management plan.”
- “How do I accept contributions from collaborators?”

That is OSPO work. The frame makes it legible to you, your institution, and the researchers you serve.

**Raise the floor, not
just solve the ticket**
Fixing their immediate
problem gets them

through the week.

Moving them from “it runs on my machine” to maintainable, licensed, citable software gets their whole lab there, and their students after them.

You don’t need a formal OSPO to do this. You need the frame.

**OSPO heuristics
already in your toolkit:**
health signals · bus
factor · governance files
· FAIR4RS · software
DMPs · contributor
conventions · community
governance

What you can do right now

If you advise on data-intensive research, you are probably already helping with code.

The network has already built the resources (Laura Langdon, UC OSPO):

- [README.md](#) - template + guide
- [LICENSE](#) - guide includes UC-approved options
- [CONTRIBUTING.md](#) - template + guide
- [CODE OF CONDUCT.md](#) - Contributor Covenant + guidance
- [CITATION.cff](#) + Zenodo release → citable, versioned DOI

Going further:

- [Growing a Community Around a Project](#) - UC OSPO
- [Sharing Research Software: Reproducible, Citable, and Discoverable](#) - Carpentries Incubator lesson (covers [pixi](#), CITATION.cff, Zenodo, discoverability)

Learn more / connect

UC OSPO Network: ucospo.net

- Education site: ucospo.net/education
- Mailing list, Slack, events calendar

Global network: curiOSS.org, 34 academic OSPOs worldwide and growing

Tim Dennis Data Science Center,
UCLA Library tdennis@library.ucla.edu



Slides: tinyurl.com/iassist2026



Handout:

tinyurl.com/iassist2026-handout



QR code linking to
tinyurl.com/iassist2026 —
slides for this presentation

tinyurl.com/iassist2026



IASSIST 2026 · Tim Dennis · UCLA Library DSC · ucospo.net

Speaker notes